📘 Lecture 31-Spanning Tree Protocol (STP) – Loop Prevention & Redundancy

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🧠 1️⃣ Introduction to STP

🌐 What is STP?

Spanning Tree Protocol (STP) is a Layer 2 protocol used to:

✅ Prevent network loops
✅ Avoid broadcast storms
✅ Maintain redundancy in networks

📌 Standard: **IEEE 802.1D**

🔁 2️⃣ Redundancy in Networks

💡 Why Redundancy is Important?

Modern networks must run:

🕐 **24/7/365** (always ON)

Even small downtime = ❌ business loss

⚠️ Problem: Poor Network Design

If network has:

❌ Single connection
❌ Single switch
❌ Single path

👉 Then **failure = network down**

This is called:

🔴 **Single Point of Failure**

✅ Better Network Design

✓ Multiple switches
✓ Multiple links
✓ Backup paths

This is called:

🟢 **Redundancy**

💻 Real Life Example

Think of **roads in a city** 🚗

- One road → traffic jam if blocked ❌
- Multiple roads → traffic reroutes ✅

Same in networking:

- Multiple paths = **backup available**

⚠️ 3️⃣ Problem with Redundancy

Redundancy creates **loops** 😨

🔥 Broadcast Storm

Ethernet frames:

❌ No TTL field
👉 Packets never expire

So broadcast traffic:

🔁 Loops forever
📈 Increases traffic
💥 Network crash

📡 Example

Scenario:

- 3 switches connected
- PC1 sends data to PC2

Flow:

1️⃣ PC1 → SW1
2️⃣ SW1 → SW2 & SW3
3️⃣ SW3 again sends back to SW1

👉 Infinite loop 🔁

💣 MAC Address Flapping

Switch learns MAC address like:

MAC → Port mapping

But in loop:

- Same MAC seen on different ports
- Switch keeps updating table

👉 This is called:

⚠️ **MAC Address Flapping**

🌳 4️⃣ Solution: STP (Spanning Tree Protocol)

STP solves loops by:

👉 Blocking some ports

🔧 How STP Works

✔ Creates **loop-free topology**
✔ Selects only ONE path
✔ Blocks redundant paths

🚦 Port States in STP

State	Description
Forwarding	Active traffic
Blocking	Disabled (backup)

🟠 Important Rule

Blocked ports:

❌ Do NOT forward data
✔ Only send/receive STP messages (BPDUs)

📡 5️⃣ BPDU (Bridge Protocol Data Unit)

STP uses **BPDU messages**.

🕐 Sent every **2 seconds**

📌 Purpose of BPDU

✔ Exchange network info
✔ Elect Root Bridge
✔ Maintain topology

⚠️ Important

Only **switches send BPDUs**

❌ PCs and routers DO NOT

🌉 6️⃣ Bridge = Switch

In STP:

👉 Switch is called **Bridge**

🧠 7️⃣ Root Bridge Concept

📌 What is Root Bridge?

👉 Main switch in network

All traffic decisions are based on it.

🏆 Root Bridge Election

Switch with:

🔻 **Lowest Bridge ID wins**

🧩 Bridge ID =

Bridge Priority + MAC Address

📊 Default Values

- Priority = **32768**
- So lowest MAC wins by default

⚠️ Important Rule

✓ Compare Priority first
✓ If same → compare MAC address

🧮 Extended System ID

Bridge Priority + VLAN ID

Example:

VLAN 1:

32768 + 1 = 32769

🔢 8️⃣ Bridge Priority Values

Can change only in steps of **4096**

Valid Values:

0, 4096, 8192, 12288, 16384, ...

🌲 9️⃣ PVST (Cisco STP)

Cisco uses:

🔹 **PVST (Per VLAN Spanning Tree)**

👉 Each VLAN runs its own STP

✔ Different root bridge per VLAN possible

🧠 1🔟 STP Working Steps

🥇 Step 1: Root Bridge Election

✔ Lowest Bridge ID wins
✔ All ports = **Designated Ports (Forwarding)**

🥈 Step 2: Root Port Selection

Each switch selects:

👉 ONE port → Root Port

Selection Rules:

1️⃣ Lowest Root Cost
2️⃣ Lowest Neighbor Bridge ID
3️⃣ Lowest Port ID

🥉 Step 3: Designated Port Selection

Each link chooses:

👉 ONE Designated Port

Other becomes:

❌ Non-Designated (Blocking)

📊 1 1️⃣ STP Cost Chart

Speed	Cost
10 Mbps	100
100 Mbps	19
1 Gbps	4
10 Gbps	2

📌 Important Rule

Cost calculated:

➡️ Only on **outgoing interfaces**
➡️ Add cost till reaching root bridge

🔌 1 2️⃣ Port Roles

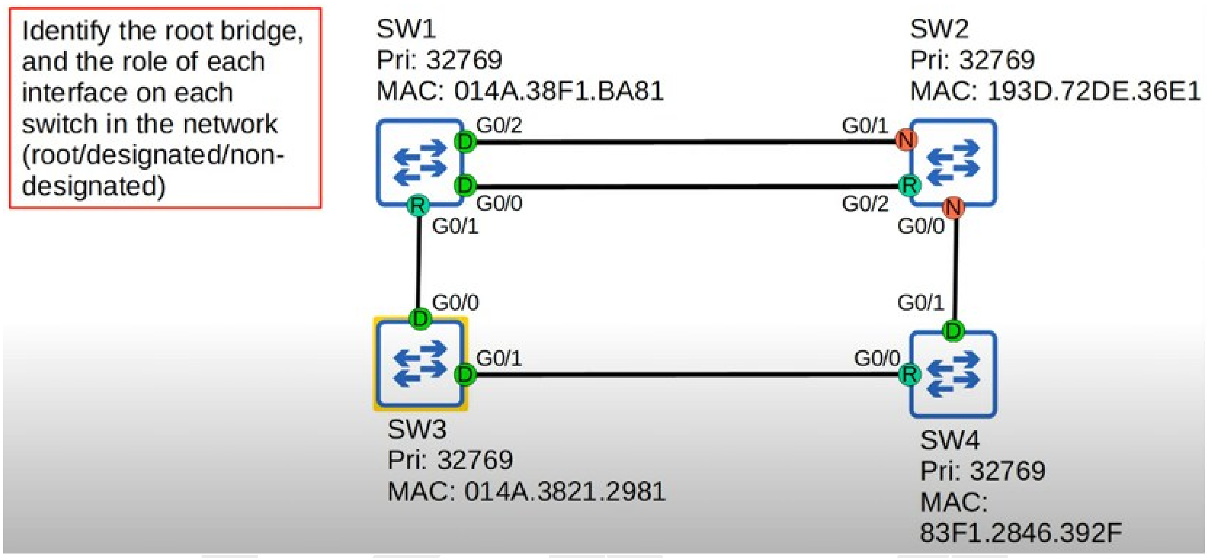
Role	Function
Root Port	Best path to root
Designated Port	Forwarding on segment
Non-Designated	Blocking

🔢 1 3️⃣ Port ID

Port ID = Priority (128 default) + Port Number

Used for tie-breaking.

🧪 1 4️⃣ QUIZ 1 – Topology Explanation



🔍 Given:

All switches have same priority → 32769

So decision based on:

👉 Lowest MAC Address

🏆 Root Bridge

✔ **SW3** (lowest MAC)

🔌 Port Roles

SW3 (Root Bridge)

✔ All ports → Designated (Forwarding)

SW1 & SW4

- Ports connecting to SW3 → Root Ports

SW2

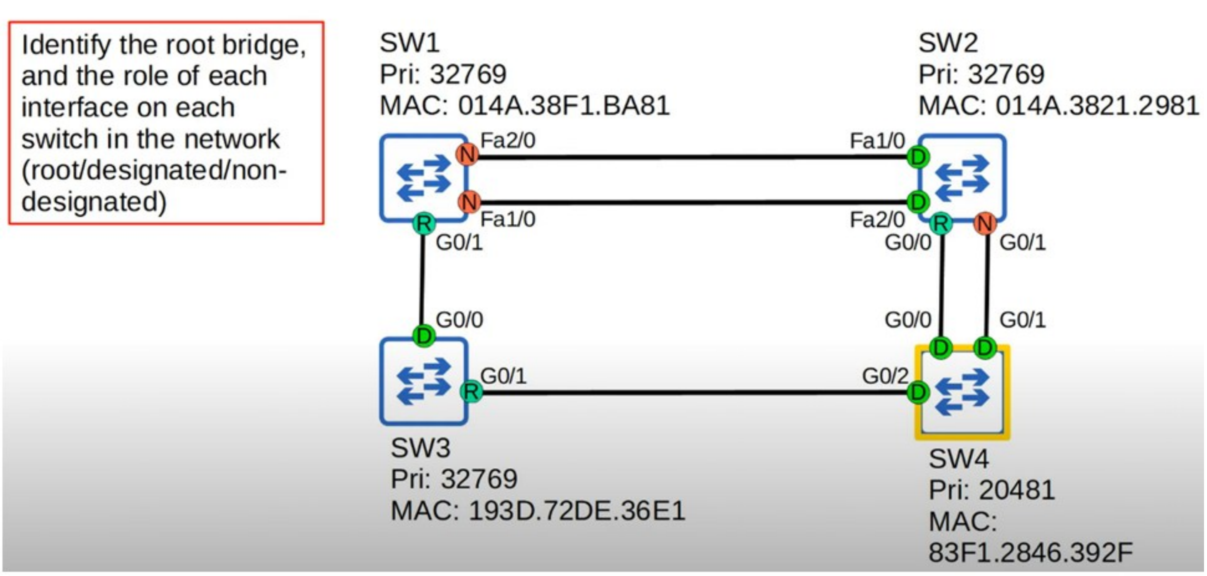
- Chooses best path to root → Root Port
- Remaining ports → Blocking

⚠️ Important Concept

If two paths exist:

👉 Lower port number wins

🧪 1 5️⃣ QUIZ 2 – Topology Explanation



🔍 Given:

SW4 priority = **20481 (lowest)**

🏆 Root Bridge

✔ **SW4**

🔌 Port Roles

SW4

✓ All ports → Designated

SW2

✓ G0/0 → Root Port (lower port number)

SW3

✓ G0/1 → Root Port

SW1

✓ G0/1 → Root Port
❌ Fa ports → Blocking (higher cost)

⚠️ Rule Used

✓ Lowest cost wins
✓ If tie → port number

🔄 1 6 STP States

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1 BLOCKING (Stable)

❌ No traffic
❌ No MAC learning
✓ Only BPDU

2️⃣ LISTENING (15 sec)

✓ Only BPDU
❌ No traffic
❌ No MAC learning

3️⃣ LEARNING (15 sec)

✓ Learns MAC
❌ No traffic

4️⃣ FORWARDING (Stable)

✓ Full traffic
✓ MAC learning
✓ BPDU

⏱️ Timers

Timer	Value
Hello	2 sec
Forward Delay	15 sec
Max Age	20 sec

🧠 Final Understanding

✅ What STP Does

✓ Prevents loops
✓ Maintains redundancy
✓ Blocks extra paths
✓ Activates backup when needed

🚀 Real Life Example

Think of **railway tracks** 🚆

- Many tracks exist (redundancy)
- Only one active track at a time
- Others are backup

STP works the same way.

🎯 Final Summary

Topic	**Key Point**
STP	Prevents Layer 2 loops
Root Bridge	Main switch
BPDU	Control message
Port Roles	Root, Designated, Blocking
States	Blocking → Listening → Learning → Forwarding
Cost	Path selection

🧠 1️⃣ Chapter Summary (Quick Revision)

🌐 What is STP?

Spanning Tree Protocol (STP) is a Layer 2 protocol used to:

✅ Prevent loops
✅ Avoid broadcast storms
✅ Maintain redundancy

📌 Standard: **IEEE 802.1D**

🔁 Why Redundancy is Needed?

Modern networks must run:

🕐 24/7/365

To avoid:

❌ Single Point of Failure

❌ Without Redundancy

One link fails → Entire network down

✅ With Redundancy

Multiple links → Backup available

⚠️ Problem: Loops in Network

When we add redundancy:

👉 Loops are created

🔥 Broadcast Storm

- Ethernet has **no TTL**

- Broadcast frames loop forever

Result:

📈 Traffic increases
💥 Network crash

💣 MAC Address Flapping

- Same MAC seen on multiple ports
- Switch keeps updating MAC table

👉 This is called **MAC Flapping**

🌳 Solution: STP

STP creates a **loop-free topology**

✓ Blocks redundant paths
✓ Keeps only one active path

🚦 STP Port States (Simplified)

State	Description
Forwarding	Active traffic
Blocking	Backup path

📡 BPDU (Important Concept)

- STP uses **BPDU (Bridge Protocol Data Unit)**
- Sent every **2 seconds**

✓ Elect root bridge
✓ Maintain topology

⚠️ Only switches send BPDUs

🌉 Root Bridge (Most Important Concept)

👉 Main switch in network

Election Rule:

Lowest **Bridge ID** wins

🧩 Bridge ID

Bridge ID = Priority + MAC Address

Default:

- Priority = 32768
- Lowest MAC wins

🔢 Bridge Priority

Valid values:

0, 4096, 8192, 12288, ...

🌲 PVST (Cisco)

👉 Per VLAN STP

✔ Each VLAN has its own STP
✔ Different root per VLAN

🧠 STP Working Steps

🥇 Step 1: Root Bridge Election

Lowest Bridge ID wins

🥈 Step 2: Root Port Selection

Each switch selects:

👉 One **Root Port**

Rules:

1️⃣ Lowest cost
2️⃣ Lowest neighbor bridge ID
3️⃣ Lowest port ID

🥉 Step 3: Designated Port Selection

Each link:

✔ One **Designated Port (Forwarding)**
❌ Others = Blocking

📊 STP Cost Table

Speed	Cost
10 Mbps	100
100 Mbps	19
1 Gbps	4
10 Gbps	2

🔌 Port Roles

Role	Meaning
Root Port	Best path to root
Designated Port	Forwarding port
Non-Designated	Blocking

🔄 STP States (Detailed)

State	Time	Description
Blocking	Stable	No traffic
Listening	15 sec	BPDU only
Learning	15 sec	MAC learning
Forwarding	Stable	Full traffic

⏱️ Timers

Timer	Value
Hello	2 sec
Forward Delay	15 sec
Max Age	20 sec

🎯 Chapter Conclusion

STP is **critical for Layer 2 network stability**.

Without STP:

❌ Loops
❌ Broadcast storms
❌ Network crash

With STP:

✔ Loop-free topology
✔ Redundancy maintained
✔ Backup links available

👉 STP intelligently blocks and activates paths to ensure **safe and efficient communication**.

🎤 Q & A

1️⃣ What is STP?

✅ Answer

Spanning Tree Protocol (STP) is a Layer 2 protocol used to prevent loops in a switched network.

It creates a loop-free topology by blocking redundant paths while still maintaining redundancy.

2️⃣ Why is STP required?

Answer

Because redundancy creates loops, which leads to:

❌ Broadcast storms
❌ MAC flapping
❌ Network instability

STP prevents these issues.

3️⃣ What is a Broadcast Storm?

Answer

A broadcast storm occurs when broadcast frames loop infinitely in the network.

Reason:

👉 Ethernet frames do not have TTL

Impact:

💥 Network congestion
💥 Switch CPU high
💥 Network crash

4️⃣ What is MAC Address Flapping?

Answer

When a switch sees the same MAC address on multiple ports repeatedly, it keeps updating the MAC table.

This is called MAC flapping and indicates a loop in the network.

5️⃣ What is BPDU?

Answer

BPDU (Bridge Protocol Data Unit) is a control message used by STP.

Functions:

✔ Root bridge election
✔ Topology maintenance

Sent every **2 seconds**

6️⃣ What is Root Bridge?

Answer

Root Bridge is the central switch in STP topology.

All path calculations are based on it.

7️⃣ How is Root Bridge elected?

Answer

Based on **lowest Bridge ID**:

1️⃣ Lowest Priority
2️⃣ If tie → Lowest MAC address

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8 What is Bridge ID?

Answer

Bridge ID = Priority + MAC Address

Used to elect root bridge.

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9 What is Root Port?

Answer

Root Port is the port on a switch that has the best path to the root bridge.

Each non-root switch has only one root port.

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10 What is Designated Port?

Answer

Designated port is the port that forwards traffic on a network segment.

Each segment has one designated port.

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1 1 What is Blocking Port?

Answer

Blocking port does not forward traffic.

It is used to prevent loops and acts as a backup.

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1 2 Explain STP states

Answer

1 Blocking – No traffic
2 Listening – BPDU exchange
3 Learning – MAC learning
4 Forwarding – Normal traffic

1️⃣3️⃣ What is STP cost?

Answer

Cost determines best path to root bridge.

Lower cost = better path

1️⃣4️⃣ What is PVST?

Answer

Per VLAN Spanning Tree (PVST) is Cisco’s implementation of STP.

Each VLAN runs its own STP instance.

1️⃣5️⃣ What happens if STP is disabled?

Answer

Network will suffer from:

💥 Broadcast storms
💥 MAC flapping
💥 Network crash

1️⃣6️⃣ How to check STP status?

show spanning-tree

1️⃣7️⃣ How to change root bridge?

spanning-tree vlan 1 priority 4096

Lower priority = higher chance to become root

1️⃣8️⃣ How to troubleshoot STP?

Steps:

✓ Check root bridge
✓ Verify port roles
✓ Check blocked ports
✓ Analyze cost

🎯 SCENARIO QUESTIONS

1️⃣9️⃣ Scenario

Network is slow and CPU is high

Answer:

Possible reason:

👉 Broadcast storm due to loop

Check:

- STP status
- Redundant links

2️⃣0️⃣ Scenario

MAC address keeps changing

Answer:

👉 MAC flapping

Reason:

Loop in network

2️⃣ 1️⃣ Scenario

Two paths available – which will STP choose?

Answer:

STP chooses path based on:

1️⃣ Lowest cost
2️⃣ Lowest bridge ID
3️⃣ Lowest port ID

2️⃣ 2️⃣ Scenario

SVI working but network unstable

Answer:

Check STP blocking ports
Possible loop issue

2️⃣ 3️⃣ What is Extended System ID?

Answer

Bridge Priority + VLAN ID

Example:

32768 + VLAN 1 = 32769

2️⃣ 4️⃣ Why STP convergence is slow?

Answer

Because of:

⏱️ Listening + Learning (30 sec total)

2️⃣ 5️⃣ What is convergence?

Answer

Time taken by network to reach stable topology after change.

🎯 One Liner

👉 “Explain STP in simple words”

Say:

“STP is a Layer 2 protocol that prevents loops in a network by blocking redundant paths while keeping backup links ready.”

🧾 FINAL REVISION POINTS

✔ STP prevents loops
✔ Root bridge is central device
✔ BPDU used for communication
✔ Ports: Root / Designated / Blocking
✔ States: Blocking → Listening → Learning → Forwarding
✔ Cost decides best path
